Health Report — Frontend Build Spec

Date: 24 March 2026
Reference samples: 7 JSONs + 7 HTML files in this folder
Validated against: all 7 HTML reference implementations

How it works

The backend outputs one file per client: `health_report_interpretations_{sample_id}.json`
The frontend reads this file and renders the full report.

Reference samples

Sample ID
1421266404096
1421029282376
1421819436544
1421425343541
1421773212865
1421093249814
1421504848853

Global colour tokens

Token	Hex	Used for
`--sand`	`#F5F0E8`	Alternating section background
`--warm`	`#FDFAF5`	Card background
`--dark`	`#1E1E2A`	Cover / driver factor
`--mid`	`#4A4858`	Body text
`--soft`	`#9A95A8`	Labels, secondary text
`--rule`	`#E4DDD0`	Borders
`--green`	`#2E8B6E`	Healthy / good
`--green-lt`	`#E8F5F1`	Green background tint
`--amber`	`#C97C2A`	Caution / moderate

Token	Hex	Used for
`--amber-lt`	`#FBF1E4`	Amber background tint
`--red`	`#C24B3A`	Critical / bad
`--red-lt`	`#FCECEA`	Red background tint
`--blue`	`#3A6EA8`	Info / research supported
`--blue-lt`	`#EAF0F8`	Blue background tint
`--purple`	`#6B5EA8`	Inferred / bidirectional
`--purple-lt`	`#F0EEF8`	Purple background tint

COVER PAGE

Score dial (large, on dark cover background)

HTML element	JSON key	Notes
Dial fill	`overall_score.total`	SVG circle r=45, circumference=283. `stroke-dashoffset = 283 × (1 - total/100)`
Dial stroke colour	`overall_score.total`	≥65 → `#2E8B6E` · ≥40 → `#C97C2A` · <40 → `#C24B3A`
Number inside dial	`overall_score.total`	
Summary text	`score_summary`	Set as `innerHTML` — contains `<strong>` tags
Key note (below summary)	`overall_score.score_drivers.key_note`	Small grey text directly below score_summary, same block

Pillar chips (5 bars below the dial)

Source: `overall_score.pillars` — always 5 keys:

Pillar key	Display name
`health_association`	Health
`diversity_resilience`	Diversity
`metabolic_function`	Metabolic
`guild_balance`	Guild Balance

Pillar key	Display name
`safety_profile`	Safety

Per chip: display name · `score / max` text · mini bar at `score/max × 100%` width
Bar colour: ≥75% → green · ≥50% → amber · <50% → red (based on `score/max × 100`)

Client profile strip (bottom of cover)

The cover-bottom strip shows stats as individual `.cover-stat` blocks:

HTML stat label	Derived from	Notes
Profile	`profile.sex` + `profile.age`	Rendered as `"{sex} · {age}"` e.g. "Male · 37" — **if `profile.sex` is null, omit this stat entirely**
Diet	`profile.diet`	Omit if `null`
Stress	`profile.stress`	Display as `N / 10`
Sleep	`profile.sleep`	Display as `N / 10`
Sensitivity	`profile.sensitivity`	
Goals	`profile.goals[]`	Array of strings · render as inline text separated by `·`

Note: `profile.first_name` is used in the HTML `<title>` tag only (`Inside Your Gut — {first_name}'s Report`), not in the visible body.

Progress banner (time slider, bottom of cover)

Source: `guild_timepoints[]`

HTML element	JSON key
Timepoint label	`guild_timepoints[i].label`
Score at timepoint	`guild_timepoints[i].score`
Guild fractions	`guild_timepoints[i].guilds` → keys: `fd bb cf bp pg md` (fractions, not %)

- `length === 1` → show "Your baseline measurement" as delta text, no slider interaction
- `length > 1` → range slider with delta vs previous timepoint

SECTION 1 — The Big Picture

Four circle score dials

Source: `circle_scores`

Key	Display title	Description
`gut_lining`	Gut Lining Protection	How well your gut wall is protected and maintained
`inflammation`	Inflammation Control	How favorable your microbiome is for keeping inflammatory pressure low
`fiber_processing`	Fiber Processing	How efficiently your bacteria ferment and process fiber
`bifidobacteria`	Bifidobacteria Presence	The abundance of your beneficial Bifidobacteria

All values 0–100. SVG circle r=37, circumference=232.
`stroke-dashoffset = 232 × (1 − score/100)`
Colour: ≥75 → green · ≥50 → amber · <50 → red

Time slider (Section 1)

Same `guild_timepoints[]` data as cover banner. Dark rounded box above the guild bars.

Guild bars

Source: `bacterial_groups` — object with 6 guild entries. **Key order varies per sample — do not hard-code order.**

Per guild bar:

HTML element	JSON key	Notes
Guild name	key name (e.g. `"Fiber Degraders"`)	`.gbar-name`
Status badge	see status table below	`.gbar-badge`
Bar fill width	`.abundance` / track max	`.gbar-fill`
Healthy range zone	hardcoded per guild	`.gbar-range` — shaded overlay on bar
Note text	`.client_interpretation`	`.gbar-note` below bar

Status → badge + colour (confirmed from HTML JS):

The HTML JavaScript explicitly maps `'below'` → `'critical'` (red) for all beneficial guilds (non-inverted). There is no blue "below" state for beneficial guilds in the rendered HTML.

Guild type	Status	Badge text	Card background	Bar colour
Beneficial	`"Within range"`	`✓ Healthy · {N}%`	`--green-lt`	green
Beneficial	`"Below range"`	`⚠ Below range · {N}%`	`--red-lt`	red (not blue)
Beneficial	`"Above range"`	`↑ High · {N}%`	`--amber-lt`	amber
Contextual	`"Within range"`	`✓ Controlled · {N}%`	`--green-lt`	green

Guild type	Status	Badge text	Card background	Bar colour
Contextual	`"Above range"`	`↑ Elevated · {N}%`	`--red-lt`	red

Beneficial guilds: Fiber Degraders, Bifidobacteria, Butyrate Producers, Cross-Feeders
Contextual guilds: Mucin Degraders, Proteolytic Guild

Reference ranges and bar track maxes:

Guild	Healthy range	Track max
Fiber Degraders	30–50%	55%
Bifidobacteria	2–10%	22%
Cross-Feeders	6–12%	30%
Butyrate Producers	10–25%	32%
Proteolytic Guild	1–5%	22%
Mucin Degraders	1–4%	14%

Bar fill %: `abundance / trackMax × 100`, capped at 100%.
Range zone position: `left = rangeMin/trackMax × 100%`, `width = (rangeMax-rangeMin)/trackMax × 100%`

Metabolic dials (4 state indicators)

Source: `metabolic_dials`

Key	Green state	Neutral state	Red state
`main_fuel`	`carb_driven`	`balanced`	`protein_driven`
`fermentation_efficiency`	`efficient`	`ok`	`sluggish`
`mucus_dependency`	`diet_fed`	`backup`	`heavy_mucus`
`putrefaction_pressure`	`scfa_dominant`	`balanced`	`protein_pressure`

Display: `.label` as text, `.state` determines colour.

Ecological metrics

Source: `ecological_metrics`

HTML element	JSON key
Shannon diversity value	`ecological_metrics.shannon`
Evenness value	`ecological_metrics.pielou_evenness`
State label	`ecological_metrics.diversity_state`

Safety banner

Source: `safety_profile`

- `any_detected === true` → show red warning banner; highlight markers where value > 0.1
- `any_detected === false` → no banner

Marker keys: `F_nucleatum` · `S_gallolyticus` · `P_anaerobius` · `E_Shigella`

SECTION 2 — Strengths & Challenges

Two-column layout

Strengths card (`.sw-card.strengths`, green background): Source: `strengths_challenges.strengths[]`

Challenges card (`.sw-card.weaknesses`, red background): Source: `strengths_challenges.challenges[]`

Per item (both cards):

HTML element	JSON key	Notes
Dot colour	`.severity`	`"critical"` / `"high"` → `dot-red` · `"moderate"` → `dot-amber` · (strengths always `dot-green`)
Emoji + title + text	`.icon`, `.title`, `.text`	Rendered as bold title + plain text inline

Expanded lists:

- `strengths_challenges.all_strengths[]` — full untruncated list
- `strengths_challenges.all_challenges[]` — full untruncated list

Use `.strengths[]` / `.challenges[]` for the initial view. Consider `all_*` for a "show more" expansion if needed.

Bottom quote block

The `.sw-bottom` renders a **static phrase** that varies per sample (e.g. "The main issue is not total collapse…" or "Despite these imbalances, the pattern is addressable…"). This text is **not** from the `good_news` JSON field.

> **Note:** The `good_news` field exists in the JSON but is not currently rendered in any of the 7 reference HTML files. Reserve this field for future use if needed, but do not render it in the current Section 2 implementation.

SECTION 3 — The Story Behind Your Results

Layout decision

```
if (root_cause_data.factor_cards.length > 0)
  → PRIMARY layout (Factor-First)
else
  → FALLBACK layout (Deviation cards only)

awareness_chips always rendered at the bottom (if non-empty)
```

Section summary block

Source: `root_cause_data.section_summary`
Green left-border highlight block (`--green-lt` background, `--green` left border).
If empty string → skip entirely (4 of 7 samples have no section summary).

PRIMARY layout — Factor-First

Used when `root_cause_data.factor_cards.length > 0` (all 7 reference samples use this layout).

Metrics strip

Source: `root_cause_data.metrics_strip[]` (up to 4 chips)
CSS class: `.ms-card`

HTML element	JSON key	Notes
Icon	`.icon`	`.ms-icon`
Label	`.client_label`	`.ms-label`
Value	`.value_str`	`.ms-value` — coloured by impact
Range	`.range_str`	`.ms-range` — prefixed "Target: "
Card class	`.impact`	`"crit"` → `.ms-crit` (red-lt) · `"high"` → `.ms-high` (amber-lt) · `"low"` → `.ms-low` (red-lt)

Value colour: `ms-crit` and `ms-low` → `--red` · `ms-high` → `--amber`

Factor cards

Source: `root_cause_data.factor_cards[]`
CSS class: `.fc-card`

Card header (`.fc-header`):

HTML element	JSON key	CSS class
Icon	`.icon`	`.fc-icon`

HTML element	JSON key	CSS class
Factor name	`.label`	`.fc-label`
Subtitle	`.subtitle`	`.fc-subtitle`
Evidence badge	`.evidence_label`	`.fc-ev-badge` + colour class
Guild dot chips	`.guild_impacts[]`	`.fc-guild-dots` → `.fc-dot`

Evidence badge colours:

`evidence_label`	CSS class
`"Well established"`	`.ev-label-established` (dark background)
`"Research supported"`	`.ev-label-supported` (blue background)
`"Emerging research"`	`.ev-label-emerging` (amber background)

Guild dot chip colours (`.fc-dot`):

`guild_impacts[j].impact`	CSS class
`"crit"`	`.fc-dot-crit` (red)
`"high"`	`.fc-dot-high` (amber)
`"low"`	`.fc-dot-low` (red)

Note: In the PRIMARY layout, `.fc-card` has **no directionality-based left border**. The `directionality` field in `factor_cards[]` is used only in the cascade diagram arrows and in the FALLBACK layout's `details.rc-factor-item` elements.

Card body (`.fc-body`):

HTML element	JSON key	Notes
Explanation text	`.explanation`	`.fc-text`
"What does science say" expand	`.kb_text`	`details.rfc-science` — **only render if non-empty string**
Guilds count	`.guilds_affected_count`	`.fc-scope` — shown as `"N / 4 guilds affected"`

Cascade diagram

CSS: `.cascade-section` / `.cascade-diag`

Left column — factor pills (`.casc-factor-pill`):

HTML element	JSON key	Notes
Emoji + label	`.icon` + `.label`	from `factor_cards[]`

HTML element	JSON key	Notes
Arrow suffix	`.directionality`	`"bidirectional"` → `↔` · anything else → `→`

Right column — guild pills (`.casc-guild-pill`):

HTML element	JSON key	CSS class
Guild name	`.client_label`	`.cgp-label`
Value	`.value_str`	`.cgp-value`
Pill colour class	`.impact`	`"crit"` → `.gp-crit` (red-lt) · `"high"` → `.gp-high` (amber-lt) · `"low"` → `.gp-low` (red-lt)
Driving emojis	`.driving_factor_emojis[]`	`.cgp-emojis` — space-separated emoji string

FALLBACK layout — Deviation cards

Used when `root_cause_data.factor_cards.length === 0`.
Also rendered inside `display:none` wrapper in PRIMARY layout for backward compat.

Source: `root_cause_data.deviation_cards[]`

Per deviation card (`.rc-dev-block`):

Deviation header — shown as plain text narrative:

HTML element	JSON key
Narrative paragraphs	`.narrative` — split on `"\n\n"` → each part as `<p>`
Fallback	`.health_meaning` — single `<p>` if `.narrative === ""`

Contributing factor chips — after narrative (`.rc-factors-label` + `.rc-factor-stack`):

Source: `.kb_drivers[]`
CSS: `details.rc-factor-chip`

HTML element	JSON key	Notes
Icon	`.icon`	`.rfc-icon`
Label	`.label`	`.rfc-title`
Evidence badge	`.evidence_label`	`.rfc-ev-label` + colour class (same as factor cards)
Body text (expanded)	`.text`	`.rfc-text`

HTML element	JSON key	Notes
"What does science say"	`.kb_text`	`details.rfc-science` — **only if non-empty**

Directionality left border — on `details.rc-factor-item` in this layout only:

`directionality` value	CSS class	Left border colour
`"driver"`	`.rfi-driver`	`--dark`
`"bidirectional"`	`.rfi-bidirectional`	`--purple`
`"consequence"`	`.rfi-consequence`	`--amber`
`"associative"`	`.rfi-associative`	`--blue`

Takeaway line (`.rc-takeaway`):

HTML element	JSON key	Notes
Green takeaway block	`.summary_line`	Prefixed `"✓ "`
Fallback	`.personal_synthesis`	If `.summary_line === ""`

If both are empty → omit the takeaway block.

Awareness chips

Source: `root_cause_data.awareness_chips[]`
If empty array → hide entire sub-section.

HTML block: `.rc-awareness-zone` with label "From your questionnaire" and intro paragraph.

CSS: `details.rc-awareness-chip`

HTML element	JSON key
Icon	`.icon` (`.rc-aw-icon`)
Label	`.domain_label` (`.rc-aw-title`)
Direction text	`.directionality_arrow` (`.rc-aw-arrow`)
Body text (expanded)	`.summary_text` (`.rc-aw-body`)

Each chip also shows a "You reported this" badge pill inline in the summary.

SECTION 4 — The Road Ahead

4-Phase timeline (left column)

Source: `timeline_phases[]` — always 4 items
CSS: `.timeline` / `.tl-item`

HTML element	JSON key	CSS class
Period label	`.weeks`	`.tl-period`
Phase title	`.title` (emoji + text)	`.tl-title`
Description	`.body`	`.tl-body`
Dot colour	`.color`	`style="background:{color}"` on `.tl-dot`

Lifestyle recommendations (right panel, dark background)

Source: `lifestyle_recommendations[]`
CSS: `.lifestyle-panel`
If empty array → show 3 hardcoded fallbacks:

Emoji	Title	Text
🥬	Increase fiber variety	"Aim for 30g of fiber daily from diverse sources…"
🏃	Stay physically active	"Regular moderate exercise (30 min, 5 days/week)…"
😴	Prioritize sleep quality	"Maintain a consistent sleep schedule and aim for 7–9 hours…"

Per item:

HTML element	JSON key	CSS class
Emoji	`.emoji`	`.lp-emoji`
Title	`.title`	`.lp-title`
Text	`.text`	`.lp-text`

SECTION 5 — Your Formula

Morning / Evening grouping

CSS: `.supp-timing-group` / `.supp-timing-banner`

Split `supplement_cards[]` into two groups based on `timing_group`:

`timing_group` value	Banner class	Banner label
`"morning"`	`.morning`	🌅 Morning
`"evening"`	`.evening`	🌙 Evening
missing/empty	infer from `.timing` string	contains "bed" or "🌙" → evening · else → morning

Important: The Magnesium card (`delivery_format_magnesium_capsule`) frequently has no `timing_group` — always infer from `.timing`.
Important: Morning Wellness Capsule card uses 🌙 in its own `emoji` field despite being morning — never use the card's `emoji` field for timing group inference; use `timing_group` only.

Render Morning group first, then Evening group. Each banner shows a count badge: "{N} units".

Supplement cards

Source: `supplement_cards[]`
CSS: `.supp-unit`

HTML element	JSON key	CSS class	Notes
Number block	`.num`	`.supp-num-block`	Background: `.color`
Card title	`.name`	`.supp-unit-name`	
Timing text	`.timing`	`.supp-unit-when`	
"Why you're taking it" band	`.why`	`.supp-why-band`	**Omit entire band if `.why === ""`**
Ingredient chips	`.pills[]`	`.pill`	`.name` bold + `·` + `.dose`
Support tags	`.supports[]`	`.sc-chip`	**Omit entire row if empty array**

> **Note on `what_it_is`:** This field exists in the JSON but is **not rendered** in any of the 7 reference HTML files. The current HTML only shows: header, why-band, pills, supports.

Pill dose rule: if dose is a bare number (no unit), append "g".

Delivery format variants

key value	Card name
`delivery_format_1_probiotic_capsule`	Probiotic Capsule
`delivery_format_2_omega_softgels`	Omega & Antioxidant Softgel
`delivery_format_3_daily_sachet`	Daily Prebiotic Sachet (includes vitamins/minerals in pills)
`delivery_format_3_powder_jar`	Daily Prebiotic Powder (fiber + botanicals only)
`delivery_format_4_evening_capsule`	Evening Capsule
`delivery_format_4_morning_wellness_capsules`	Morning Wellness Capsule
`delivery_format_5_evening_wellness_capsules`	Evening Wellness Capsule

key value	Card name
`delivery_format_6_polyphenol_capsule`	Polyphenol Capsule — if `timing_group === "morning"` → show amber warning: "Take with breakfast or on full stomach."
`delivery_format_magnesium_capsule`	Magnesium Bisglycinate — no `timing_group`, always infer from `.timing`

SECTION 6 — Your Goals

Source: `goal_cards[]` — 3 or 4 items
CSS: `.goal-card`

HTML element	JSON key	CSS class	Notes
Emoji	`.emoji`	`.goal-emoji`	
"Also addressed" badge	`.inferred`	`.goal-inferred-tag`	Show only if `true` — rendered before `.goal-title`
Title	`.title`	`.goal-title`	
Mechanism text	`.mechanism`	`.goal-mechanism`	
Formula box	`.formula_link`	`.goal-formula`	Purple left-border box, prefixed `<strong>Your formula:</strong>`

REFERENCES

Source: `cited_papers[]` (top-level key — not `root_cause_data.cited_papers`)
If empty array → hide entire References section.

Format as APA: `Surname, I. (year). Title. *Journal*. https://doi.org/…`
DOI as clickable `<a>` link. Journal in italics.

FOOTER

CSS: `.footer`

HTML element	Source
Brand	Static: "NB1 Health" (`.footer-brand`)
Disclaimer	Static: "This report is for informational purposes only and does not constitute medical advice…"

HTML element	Source
Date	`report_date` → parse `YYYY-MM-DD` → display as `DD Month YYYY`